

Table of Contents

[VHFACTriggerControlDefTopic](#)
[VhfacTriggerControl_about.1.2](#)
[VhfacTriggerControl_debug.4.1](#)
[VhfacTriggerControl_debug.4.2](#)
[VhfacTriggerControl_debug.4.3](#)
[VhfacTriggerControl_prog.3.1](#)
[VhfacTriggerControl_prog.3.2](#)
[VhfacTriggerControl_prog.3.3](#)
[VhfacTriggerControl_prog.3.4](#)
[VhfacTriggerControl_theory](#)

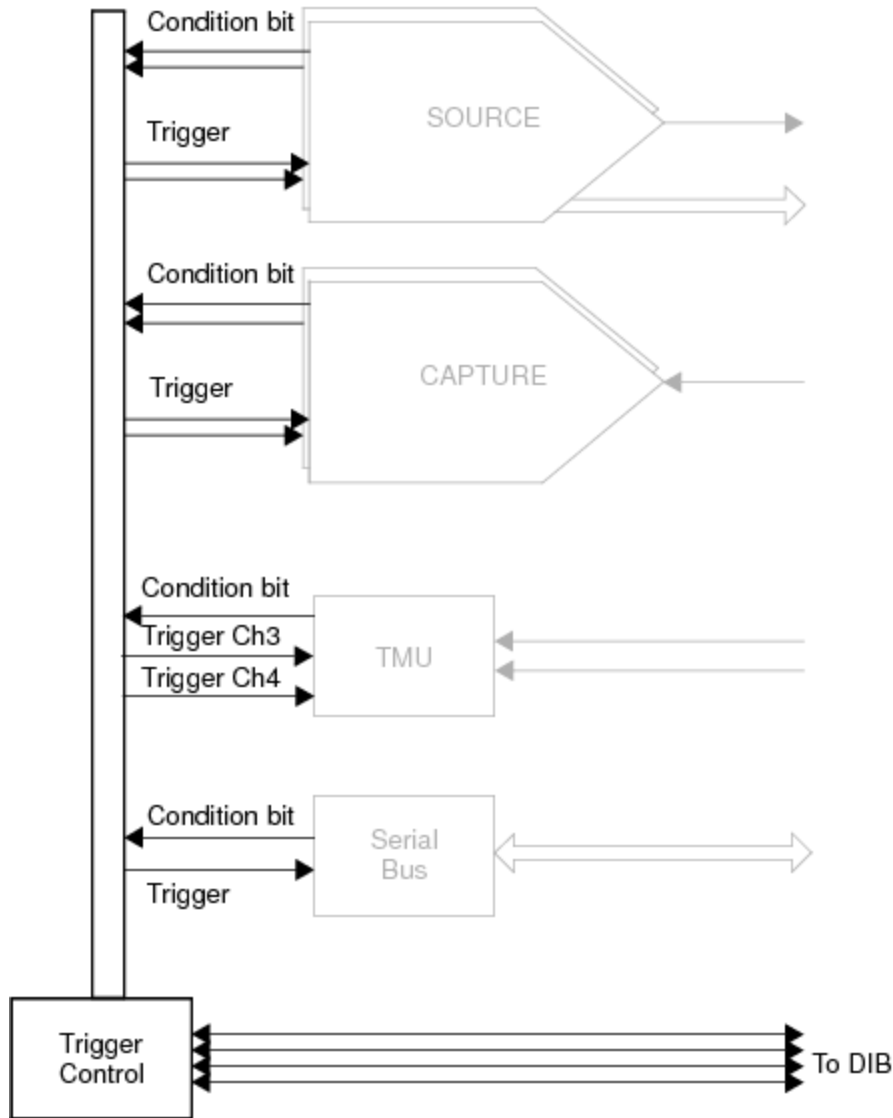
VHFACTriggerControlDefTopic



About the VHFAC Trigger Control

[About the VHFAC Trigger Control](#)

The VHFAC Trigger Control supports the operation of the instruments on the board.



VHFAC Trigger Control

This section includes the following topic:

- [Features](#)

VHFACTime information includes the following sections:

- [VHFAC Trigger Control Theory of Operation](#)
- [VHFAC Trigger Control Programming](#)
- [VHFAC Trigger Control Debug Displays](#)

Related Topics

- [VHFAC Specifications](#)
- [VHFACTrigControl](#) (VBT syntax)
- [VHFAC DIB Design Guidelines](#)

For information on other VHFAC instruments see:

- [About the VHFAC Capture](#)
- [About the VHFAC Serial Bus](#)
- [About the VHFAC Source](#)
- [About the Time Measurement Unit](#)





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About the VHFAC Trigger Control : Features

Features

The features of Trigger control include:

- | | |
|---|--|
| • | Four trigger I/O lines to DIB |
| • | Access to all instruments |
| • | Ability for instruments to trigger other instruments and the DIB |

For VHFAC Trigger Control hardware specifications, see [VHFAC Specifications](#).

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VHFAC Trigger Control Debug Displays

VHFAC Trigger Control Debug Displays

The VHFAC Trigger Control Instrument Debug Displays are interactive tools that show the state of the VHFAC Trigger instrument in real time. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a drop-down menu. Non-interactive features are grayed out on the display and cannot be adjusted during debugging. This section describes features specific to the following VHFAC Trigger Control debug displays:

- [VHFAC Trigger Control Display](#)

- [VHFAC Trigger Connect Display](#)

For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)

- [Producing Code from Debug Displays](#)

Starting VHFAC Trigger Control Instrument Debug Displays

To launch the VHFAC Trigger Control Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Trig Control.

To launch the VHFAC Trigger Connection Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Trig Connect.

To start debugging, select a pin and a site to display the programmed values. Select sites and pins from the drop-down menu.

Without a test program loaded, the initial debug display returns the following error message:

No pin exists of type VHFAC.
and the connection lines appear directly connected to the DUT (without relays, DIB Access, or PPMU instruments active). Load a validated test program and restart the debug display to remedy this situation.

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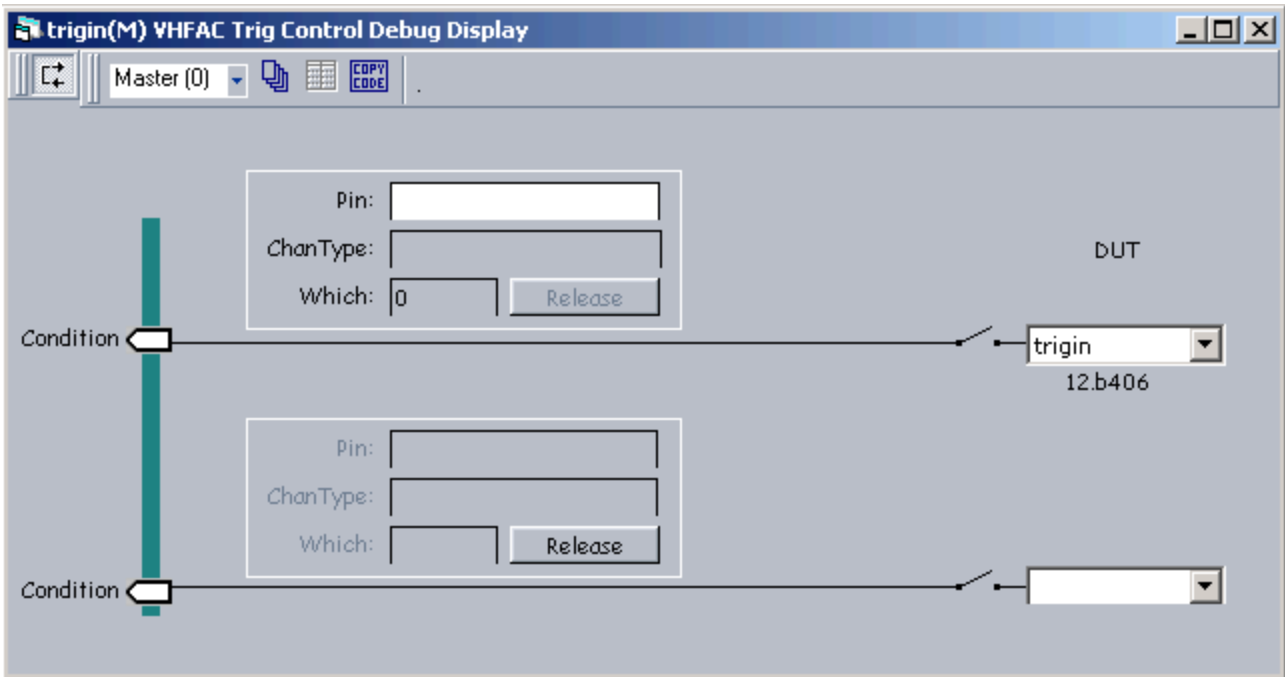
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VHFAC Trigger Control Debug Displays : VHFAC Trigger Control Display

VHFAC Trigger Control Display

The VHFAC Trigger Control Display shows the state of the four DIB trigger I/O pins. This display is only needed when the one or more of the DIB trigger resources is used to trigger an instrument on the board (not when instruments trigger one another internally, without the use of these DIB I/O pins).



VHFAC Trigger Control Debug Display

The following table lists the controls on the display.

VHFAC Trigger Control Parameters and Buttons

Parameter/Button	Description
------------------	-------------

Pin	Specifies the instrument whose condition signal is used as the trigger signal.
ChanType	The instrument channel type.
Which	Auxiliary input of the instrument.
DUT	Pin name of the DIB Trigger I/O pin.
Release	Performs a release input action on the specified DIB Trigger I/O pin.

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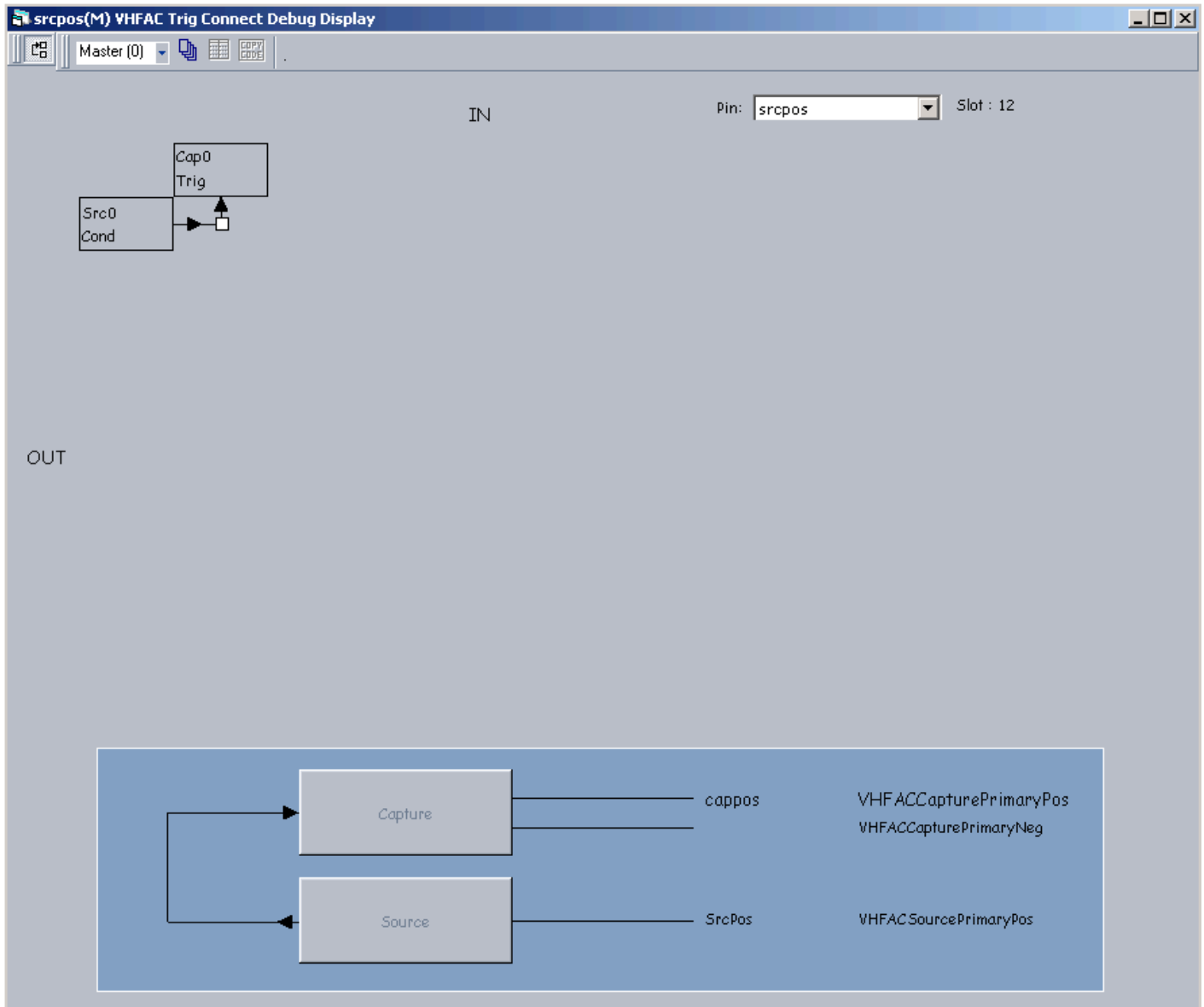
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[VHFAC Trigger Control Debug Displays](#) : VHFAC Trigger Connect Display

VHFAC Trigger Connect Display

The VHFAC Trigger Connect Debug Display shows the connection between instruments. The display shows which instrument's trigger output signals (OUT) are connected to which instrument's trigger input (IN).



VHFAC Trigger Connect Display

Click the white connection box between the instruments shown in the upper left hand part of the display to see a more detailed view of that connection. The detailed view becomes visible at the bottom of the display.

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VHFAC Trigger Control Programming

VHFAC Trigger Control Programming

This section describes programming of the VHFAC Trigger Control instrument.
The following topics are available:

- | | |
|---|--|
| • | Programming the Trigger Control Instrument |
| • | Programming the Channel Map |

For more information, see [VHFACTrigControl](#) in the VBT Syntax Reference.

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VHFAC Trigger Control Programming : Programming the Trigger Control Instrument

Programming the Trigger Control Instrument

The main steps to programming the Trigger Control instrument are:

1. Set up the instrument that will generate the trigger signal.
2. Set up the instrument that will receive, and be triggered by, an external trigger signal.
3. Set up the DIB trigger I/O pins.

However, not all steps apply to all test cases.

Set Up the Instrument That Will Generate the Trigger Signal

Set up the instrument condition bit. For information on how to program the condition bit, see:

- VHFAC Capture Condition Bit
- VHFAC Source Condition Bit

Set Up the Instrument That Will Receive and Be Triggered by an External Trigger Signal

1. Program the origin of the trigger signal. Either:
 - a. Another instrument on the same board
 - b. A DIB Trigger I/O pin on the same board

VBT Example:

```
TheHdw.VHFACCApture.Pins("CapPin").AsynchronousTrigger.SetInput "SrcPin"
```

2. Set the Microcode control to an External Trigger mode.

VBT Example:

```
TheHdw.VHFACCApture.Pins("CapPin").Signals("CapSig").Item. _  
MicroCodeControl = tlVHFACCAptureMicrocodeControlByExternalTrigger1
```

3. Create and define the instrument's (the one being triggered) "External Trigger Setup"

- a. Specify the action to occur when the trigger is received.

- b. Load the settings.

- c. Assign a "Start Label".

VBT Example:

```
With TheHdw.VHFACCApture.Pins("CapPin")  
.ExternalTriggers.Item("EXT_TRG").Add tlVHFACCAptureCommandTRIG, "CapSig"  
.ExternalTriggers.Load ("EXT_TRG")  
.ExternalTriggers.StartLabel = "EXT_TRG"  
End With  
Set Up the DIB Trigger I/O Pins
```

1. Program the origin of the trigger signal.

VBT Example:

```
TheHdw.VHFACTrigControl.Pins("TrigOutPin").SetInput "SrcPin"
```

2. Connect the pins to the DIB.

VBT Example:

```
TheHdw.VHFACTrigControl.Pins("TrigOutPin").Connect
```



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VHFAC Trigger Control Programming : Programming the Trigger Control Instrument : Programming Example

Programming Example

This programming example uses the steps described above. The condition bit in this example comes from the Source instrument and the Capture instrument is the instrument which receives the trigger. The trigger signal is routed from the Source to a DIB trigger I/O pin. Once the trigger signal reaches the DIB, it is then routed to a second DIB trigger I/O pin with a jumper or DIB trace. The trigger signal on this second DIB trigger I/O pin is routed to the Capture asynchronous trigger input.

VBT Example

' Set up the Source Condition Bit

With TheHdw.VHFACSource.Pins(SrcPin).Condition

.Connect tIVHFACSourceConditionToBoard

.Enable = True

End With

' Set up Trigger Matrix

TheHdw.VHFACCapture.Pins(CapPin).Signals.Add ("CapSig2")

With TheHdw.VHFACCapture.Pins(CapPin)

.AsynchronousTrigger.SetInput TrigInPin

.ExternalTriggers.Item("EXT_TRG").Add tIVHFACCaptureCommandTRIG, "CapSig2"

.ExternalTriggers.Load ("EXT_TRG")

.ExternalTriggers.StartLabel = "EXT_TRG"

End With

```
TheHdw.VHFACTrigControl.Pins(TrigOutPin).SetInput SrcPin
TheHdw.VHFACTrigControl.Pins(TrigOutPin).Connect
TheHdw.VHFACTrigControl.Pins(TrigInPin).Connect
```

' Set up Capture

```
With TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig2").Item
    .SampleSize = 16384
    .SampleRate = 80000000#
    .MicroCodeControl = tlVHFACCaptureMicrocodeControlByExternalTrigger1
    .LoadSettings
End With
```

' Start Source (which also triggers capture via source condition bit and the
' trigger matrix)

```
TheHdw.VHFACSource.Pins(SrcPin).Signals("SrcSig2").Start tlStartImmediately
```

Comments

In this example, the VBT code:

```
With TheHdw.VHFACSource.Pins(SrcPin).Condition
    .Connect tlVHFACSourceConditionToBoard
    .Enable = True
End With
```

sets up the source condition bit as the trigger signal.

The VHFAC capture trigger input is set to TrigInPin:

```
.AsynchronousTrigger.SetInput TrigInPin
```

These lines define the VHFAC capture and external trigger setup:

```
.ExternalTriggers.Item("EXT_TRG").Add tlVHFACCaptureCommandTRIG, "CapSig2"
.ExternalTriggers.Load ("EXT_TRG")
.ExternalTriggers.StartLabel = "EXT_TRG"
```

The Source condition bit is routed to the "TrigOutPin" to be sent to the DIB in this line:

```
TheHdw.VHFACTrigControl.Pins(TrigOutPin).SetInput SrcPin
```

Both the trigger input and output signal pins are connected to the DIB with these connect statements:

```
TheHdw.VHFACTrigControl.Pins(TrigOutPin).Connect
TheHdw.VHFACTrigControl.Pins(TrigInPin).Connect
```

This line sets up the VHFAC capture to be triggered via the Trigger Control Instrument:

```
.MicroCodeControl = tlVHFACCaptureMicrocodeControlByExternalTrigger1
```


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VHFAC Trigger Control Programming : Programming the Channel Map

Programming the Channel Map

In creating your channel map, make sure the channel type resources and Signal Names are correct. Select a channel type from the Type column drop-down list for each pin. Valid VHFAC Trigger Control signal types are:

VHFACTrigControl0	VHFAC Trigger Control 0 (2 trigger Lines)
VHFACTrigControl1	VHFAC Trigger Control 1 (2 trigger Lines)

Type the tester channel information value for each site to connect a DUT pin to the instrument. This table lists the available VHFAC Trigger Control channels. Note that in the table, x indicates the DIB slot number of the VHFAC Trigger channel.

VHFAC Capture Channel Resources

VHFAC Capture Pogo Pin	DIB Channel	Signal Name
VHFAC Trigger Control 0 Line 1	x.a37	x.trig0_0
VHFAC Trigger Control 0 Line 2	x.b36	x.trig0_1
VHFAC Trigger Control 1 Line 1	x.c37	x.trig1_0
VHFAC Trigger Control 1 Line 2	x.d36	x.trig1_1

For more information, see [Channel Map Sheet](#) in the DataTool section.

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VhfacTriggerControl_theory



VHFAC Trigger Control Theory of Operation

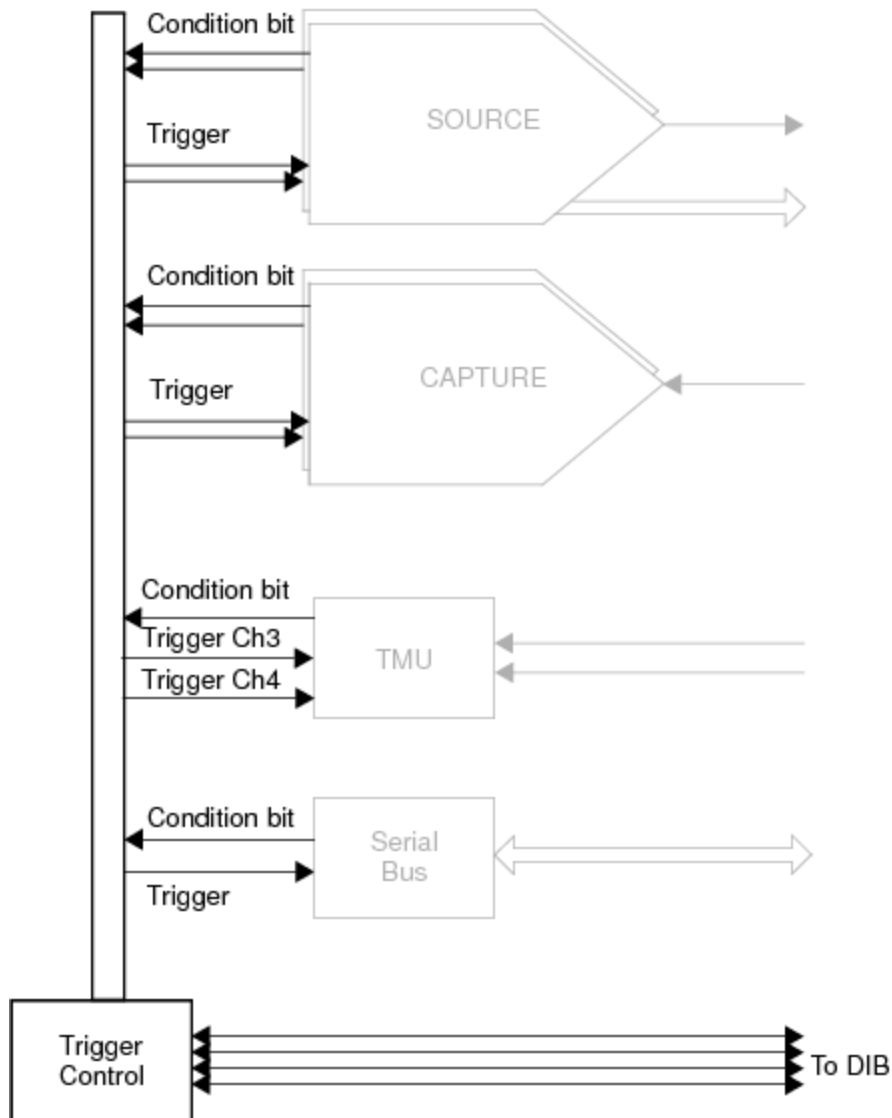
VHFAC Trigger Control Theory of Operation

In general, the VHFAC Trigger Control Instrument is used to distribute “trigger” signals between VHFAC instruments and/or the DIB.

Specifically, the VHFAC Trigger Control Instrument allows:

- A condition bit signal from one instrument to be used as a Trigger signal for another instrument internally, on the (same) board.
- A condition bit signal from an instrument to be used as an output signal to the DIB using one or multiple trigger I/O pins.
- An input signal from the DIB to be used as the Trigger signal for an instrument on the (same) board using one of the trigger I/O pins.

This instrument is useful for applications such as video device testing, where it is often necessary to synchronize multiple source and capture instruments, and may be difficult to achieve pattern synchronization. The Trigger Control instrument can be used to distribute a condition signal generated by a VHFAC Source instrument to Capture instrument on a different board. VHFAC instruments are internally connected using internal trigger lines. The following diagram shows the trigger control connections:



Trigger Control Connections

For information on VHFAC Trigger Control instrument performance specifications and standards, see [VHFAC Specifications](#).

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